

Nonlinear Kicker Magnet (NKM), BTS Transfer Line Optical Matching, and Storage Ring Injection Dynamics

Consolidated Technical Report, Literature Review, and Simulation Manual

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Abstract

This consolidated technical document presents a comprehensive theoretical framework, numerical modeling strategy, software architecture, and publication reproduction workflow for the Nonlinear Kicker Magnet (NKM) system and the Booster-to-Storage Ring (BTS) transfer line optics at the 4th Generation Storage Ring (4GSR). The primary objective of the NKM is to enable transparent off-axis top-up beam injection into the storage ring by generating a octupole-like non-uniform horizontal magnetic kick that vanishes on the closed orbit ($x = 0$), thereby eliminating stored-beam perturbation while providing sufficient deflection to capture injected electron bunches. This report consolidates all analytical physics models, 2D/3D RADIA magnetic field calculation ingestion algorithms, thick symplectic particle tracking formulations, multi-turn physical aperture storage ring tracking dynamics, multi-objective genetic algorithm (MOGA) optics optimization, and cryptographic data provenance mechanisms into a unified reference document.

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1 Physics Models & Theoretical Foundations

1.1 Overview of Off-Axis Storage Ring Beam Injection

Modern high-brilliance synchrotron radiation sources require top-up injection to maintain constant stored beam current and thermal stability of beamline optics. Traditional injection schemes employ conventional dipole bump kickers, which create transient orbit distortions for the stored circulating beam due to kicker pulse mismatch and finite pulse widths. The Nonlinear Kicker Magnet (NKM), also referred to as a Nonlinear Kicker (NLK), replaces pulsed dipole bumps with a single pulsed non-axisymmetric magnetic field configuration.

The ideal NKM magnetic field profile along the horizontal midplane ($y = 0$) exhibits a zero field and zero spatial gradient on the reference orbit ($x = 0$), while producing a peak transverse deflection at an offset corresponding to the injected beam position ($x \approx -16$ mm to -18 mm). Consequently, circulating electron bunches passing through the center of the NKM experience zero kick, eliminating residual stored-beam centroid oscillations and transverse emittance growth.

1.2 Analytical 4-Wire NLK Field Model

The classical analytical model for an NKM consists of four parallel long current-carrying conductors positioned symmetrically around the vacuum chamber at coordinates $(\pm x_c, \pm y_c)$. Carrying equal current magnitudes I with alternating polarities, the magnetic flux density $\vec{B}(x, y) = B_x \hat{x} + B_y \hat{y}$ on the transverse plane is derived from the Biot-Savart law via complex potential techniques:

$$B_y(x, y) = \frac{\mu_0 I}{2\pi} \sum_{k=1}^4 (-1)^k \frac{x - x_k}{(x - x_k)^2 + (y - y_k)^2} \quad (1)$$

For an ultra-relativistic electron bunch with rest charge $q = -e$ and nominal momentum $p_0 = \gamma m_e v_0$, the net horizontal deflection angle $\Delta x'$ experienced across an active magnet length L_{nkm} is expressed by the integrated Lorentz force equation:

$$\Delta x' = \frac{q}{p_0} \int_0^{L_{\text{nkm}}} B_y(x, y, z) dz = \frac{1}{(B\rho)_0} \int_0^{L_{\text{nkm}}} B_y(x, y, z) dz \quad (2)$$

where $(B\rho)_0 = \frac{p_0}{q}$ is the magnetic rigidity of the electron beam (13.3426 T · m at 4.0 GeV). The sign convention is defined such that a negative horizontal offset $x < 0$ yields a positive angular kick $\Delta x' > 0$, directing the injected beam towards the storage ring reference orbit.

1.3 3D & 2D RADIA Magnetic Field Maps

While the analytical 4-wire model offers qualitative insight, realistic NKM hardware features finite pole geometries, conductor terminations, and 3D fringe fields. 3D magnetic field calculations performed in RADIA produce explicit field arrays:

1. **1D On-Axis Field Profile** ($B_y(z)$): Tabulated along the longitudinal axis ($z \in [-0.5 \text{ m}, +0.5 \text{ m}]$) to evaluate peak integrated magnetic field strength $J_y = \int B_y dz \approx 0.040 \text{ T} \cdot \text{m}$.
2. **2D Transverse Kick Map** ($K_x(x, y)$): Tabulated integrated horizontal kick angles $\Delta x'(x, y)$ over a 2D spatial grid covering $x \in [-30 \text{ mm}, +30 \text{ mm}]$ and $y \in [-15 \text{ mm}, +15 \text{ mm}]$.

To evaluate kicks at arbitrary particle coordinates during tracking without introducing unphysical artificial numerical noise, 2D bicubic spline interpolation with strict domain boundary checks is implemented.

1.4 Thick Symplectic Particle Tracking

To preserve phase-space volume and numerical stability over multi-turn simulations, the NKM element is divided into N_{slices} thick integration steps. Each slice of length $\Delta z = L_{\text{nkM}}/N_{\text{slices}}$ is evaluated using a second-order split-operator symplectic integrator:

$$x_{k+1/2} = x_k + \frac{1}{2}\Delta z x'_k \quad (3)$$

$$x'_{k+1} = x'_k + \frac{q}{p_0} B_y(x_{k+1/2}, y_{k+1/2}) \Delta z \quad (4)$$

$$x_{k+1} = x_{k+1/2} + \frac{1}{2}\Delta z x'_{k+1} \quad (5)$$

Tracking convergence studies confirm that $N_{\text{slices}} \geq 20$ guarantees phase-space angular convergence to within $< 10^{-6}$ rad.

1.5 Multi-Turn Storage Ring Tracking & Physical Apertures

The storage ring lattice consists of 3,483 elements modeled using the Accelerator Toolbox (**pyAT**). The NKM element is inserted into the first straight section downstream of the injection septum. Injected particles enter the ring at an offset of $x_{\text{inj}} = -16$ mm with initial angular tilt $x'_{\text{inj}} = +3.0$ mrad.

Physical apertures of ± 30 mm (horizontal) and ± 15 mm (vertical) are applied at every lattice element across $N_{\text{turns}} = 1000$ turns. Particles exceeding these spatial boundaries are flagged as lost, and turn-by-turn transmission efficiency η_{cap} is computed.

2 Literature Review & Context

2.1 Historical Evolution of Storage Ring Kicker Systems

Conventional injection into electron storage rings relies on a 4-kicker local bump, where four identical pulsed dipole magnets create a temporary closed orbit distortion. However, residual pulse mismatches between kickers generate orbit noise on the order of several beam sizes (σ_x), disrupting sensitive user experiments.

To overcome this, non-axisymmetric kickers were pioneered:

- **BESSY II (Helmholtz-Zentrum Berlin)**: First operational demonstration of a 4-rod Nonlinear Kicker magnet, achieving top-up injection without measurable stored-beam disruption.
- **MAX IV (Lund University)**: Implementation of a pulsed octupole injection magnet in a ultra-low emittance multi-bend achromat (MBA) storage ring.
- **SOLEIL & ALS-U**: Adoption of optimized NLK profiles to accommodate narrow vacuum chamber stay-clears and off-axis top-up injection.

2.2 4GSR Injection Requirements

The 4th Generation Storage Ring (4GSR) is a 4.0 GeV ultra-low emittance facility ($e_{\text{nat}} \approx 50$ pm · rad). The BTS transfer line transports 4.0 GeV electron bunches from the booster synchrotron to the storage ring injection septum.

Optical matching at the septum exit requires precise control of Twiss parameters ($\beta_x = 7.56$ m, $\beta_y = 12.27$ m, $\alpha_x = 1.52$, $\alpha_y = -1.65$) and zero dispersion to maximize injection transmission efficiency while constraining maximum quadrupole strengths within hardware limits ($|K_1| \leq 3.0$ m⁻²).

3 Simulation Software Architecture

The software framework is structured under the `src/nkm/` package, following modular object-oriented principles with strict unit enforcement:

Table 1: `src/nkm/` Module Subsystem Architecture

Module	Primary Purpose	Key Classes / Functions
<code>units.py</code>	Unit definitions & rigidity	<code>compute_rigidity()</code> , <code>integrated_field_to_kick()</code>
<code>beam.py</code>	6D particle beam distributions	<code>generate_6d_beam()</code> , <code>compute_beam_statistics()</code>
<code>bts_lattice.py</code>	36-element BTS transfer line	<code>build_bts_lattice()</code> , <code>get_bts_quad_names()</code>
<code>fieldmap.py</code>	1D longitudinal field maps	<code>NKMFieldMap1D</code> , <code>validate_1d_fieldmap()</code>
<code>kickmap.py</code>	2D horizontal kick maps	<code>NKMKickMap2D</code> , <code>verify_grid_interpolation()</code>
<code>tracking.py</code>	Symplectic tracking integrators	<code>track_nkm_thick_symplectic()</code>
<code>storage_ring_injection.py</code>	Storage ring multi-turn tracking	<code>load_storage_ring_injection_lattice()</code>
<code>optimization.py</code>	SLSQP quad matching	<code>optimize_bts_matching()</code>
<code>moga.py</code>	Multi-objective optimization	<code>run_moga_optimization()</code>
<code>errors.py</code>	Misalignment & gradient errors	<code>apply_quad_errors()</code>
<code>results_schema.py</code>	Provenance & cryptographic hashes	<code>PaperResultSchema</code> , <code>compute_input_hashes()</code>

4 Authoritative Simulation Procedures

The simulation suite comprises six sequential, reproducible workflow steps:

4.1 Step 1: NKM Field & Kick Map Cross-Validation

5-way cross-validation between RADIA 3D calculations, RADIA 2D kickmaps, 4-wire analytical models, polynomial fits, and Lorentz sign conventions. Verification checks confirm field symmetry residuals $< 10^{-4}$ and monotonicity along the horizontal axis.

4.2 Step 2: Symplectic Slicing Convergence

Convergence study evaluating thick tracking integration over $N_{\text{slices}} \in \{10, 20, 50, 100\}$. Numerical kick error decays as $\mathcal{O}(N_{\text{slices}}^{-2})$, demonstrating $< 10^{-6}$ rad residual error for $N_{\text{slices}} \geq 20$.

4.3 Step 3: Multi-Turn Storage Ring Injection Dynamics

Simulates 1,000 particles over 1,000 turns across four kicker configurations:

1. **NKM Off:** Reference baseline tracking with no kicker deflection.
2. **Ideal Kicker:** Thin-kick dipole approximation at septum offset.
3. **Linearized NKM:** Linear Taylor expansion about $x = -16$ mm.
4. **RADIA Fieldmap NKM:** Full 2D spline interpolated kick map.

Physical aperture tracking demonstrates $> 99.8\%$ injection capture efficiency and < 0.08 mm residual stored-beam centroid perturbation.

4.4 Step 4: Deterministic BTS Quadrupole Matching

2-stage SLSQP optimization adjusting 8 BTS quadrupole families to achieve target Twiss parameters at the septum. SVD Jacobian decomposition quantifies parameter sensitivity and condition numbers ($C_N \approx 42.1$).

4.5 Step 5: Monte Carlo Tolerance Budget & Sensitivity Analysis

Evaluates 500 Monte Carlo seeds with random quadrupole misalignments ($\sigma_{x,y} = 100 \mu\text{m}$), roll errors ($\sigma_\phi = 0.5 \text{ mrad}$), relative gradient errors ($\sigma_K/K = 10^{-3}$), and NKM field scaling errors ($\pm 2\%$). Results establish a 95% confidence interval for transmission efficiency ($> 98.5\%$).

4.6 Step 6: Multi-Objective Pareto Optimization (MOGA)

NSGA-II optimization balancing competing objectives:

$$\min_{\vec{K}} \left\{ f_1(\vec{K}) = M_x + M_y, \quad f_2(\vec{K}) = \max \beta_{x,y}, \quad f_3(\vec{K}) = \Delta A_{\text{stored}} \right\} \quad (6)$$

Generates a clear Pareto front detailing tradeoffs between optical mismatch and aperture stay-clears.

5 Analysis Methods & Verification Metrics

5.1 Optical Mismatch Factor

The Twiss parameter optical mismatch factor M is defined as:

$$M = \frac{1}{2} [\beta_0 \gamma + \beta \gamma_0 - 2\alpha_0 \alpha] \quad (7)$$

where $(\beta_0, \alpha_0, \gamma_0)$ are target Twiss parameters and (β, α, γ) are actual propagated values. A value of $M = 1.00$ indicates perfect optical matching, while $M \leq 1.05$ defines the acceptable tolerance window.

5.2 Cryptographic Provenance Verification

All scientific input data files are protected and cataloged via SHA-256 cryptographic hashes:

- By.txt: fa7be11ac01ab09826c997a7855050aa533c9ad5a2478684463dd9afaabcc305
- kickmap_file.txt: 5c1a3f1437cec79917515eb13443fc176550b9040553811b644db02412c2e42b
- K4GSR.HBIv4-1.mat: MAD-X imported storage ring lattice

6 References & Bibliography

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